KHYATI KIYAWAT

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Research Focus

Interests: Memory-Centric Architecture, Energy- and Cost-Efficient LLM Inference Acceleration

My research addresses memory capacity and bandwidth bottlenecks in LLM inference. I co-design processing-in-memory (PIM) and processing-near-memory (PNM) architectures, including chiplet-based DRAM and computational CXL devices, and develop performance modeling frameworks to evaluate and guide their design.

Education

University of Virginia, VA, USA 2022–2028 (expected)
Ph.D., Department of Computer Science (Advised by Prof. Kevin Skadron) GPA: 3.88/4.0

Indian Institute of Technology Roorkee, India 2016–2020
B.Tech., Department of Electronics and Communication GPA: 9.23/10.0

Industry Experience

Memory Solution Research Intern | Samsung Semiconductors, USA May 2026–Aug 2026
Mentors: Gongjin Sun, Caroline Kahn

- Characterized the performance of Samsung's CXL-based memory devices as a KV cache offload tier for LLM inference using LMCache and compared them against System DDR and SSD baselines. (US patent to be filed)

Processor Architecture Research Intern | Intel, India May 2024–Jan 2025
Mentors: Shankar Balachandran, Anant V. Nori, Sreenivas Subramoney

- Built a tensor-tracking framework for GenAI workloads that enables rapid exploration of lossless KV cache compression in Xeon LLCs.

Front-end Design Verification Engineer | Texas Instruments, India Summer 2019, Jul 2020–Jul 2022

- Verified SoC integration of registers, communication peripherals, and co-processors for C2000 real-time MCUs.
- Developed a Perl-based automation framework to auto-generate RTL for memory-mapped registers and the verification test cases, significantly cutting manual effort and register-RTL errors by 80%.

Publications

[9] **Trace-Driven Analysis for KV Cache Tier Design**
Khyati Kiyawat, Kevin Skadron. *IEEE CAL, 2026 (under review).* [link]

[8] **Reimagining LLM Inference Infrastructure with Memory-Centric KV Cache Servers**
Khyati Kiyawat, Kevin Skadron. *HotInfra, 2026 (co-located with ISCA 2026).* [link]

[7] **HARMONI: Hierarchical ARchitecture MOdeling for LLMs with Near/In Memory Computing**
Khyati Kiyawat, Yasas Seneviratne, Zhenxing Fan, Morteza Baradaran, Kevin Skadron.
ISPASS, 2026. [link]

[6] **Sangam: A Chiplet-Based DRAM-PIM Accelerator with CXL Integration for LLM Inferencing**
Khyati Kiyawat, Zhenxing Fan, Yasas Seneviratne, Morteza Baradaran, Akhil Shekar, Zihan Xia, Mingu Kang, Kevin Skadron. *arXiv preprint, Nov 2025 (under review).* [link]

[5] **TriPIM: Exact Triangle Counting on UPMEM PIM for Graph Analytics**
Morteza Baradaran, **Khyati Kiyawat**, Akhil Shekar, Abdullah Mughrabi, Kevin Skadron.
LBR MemSys, 2025. [link]

[4] **Membrane: Accelerating Database Analytics with DRAM-Based PIM Filtering and Schema Denormalization**
Akhil Shekar, Kevin Gaffney, Martin Prammer, **Khyati Kiyawat**, Lingxi Wu, Helena Caminal, Zhenxing Fan, Yimin Gao, Ashish Venkat, José F. Martínez, Jignesh Patel, Kevin Skadron. *TACO, 2025.* [link]

[3] **Architectural Modeling and Benchmarking for Digital DRAM PIM**
F. Siddique, D. Guo, Z. Fan, M. Gholamrezaei, M. Baradaran, A. Ahmed, H. Abbot, K. Durrer, K. Nandagopal, E. Ermovick, **K. Kiyawat**, B. Gul, A. Mughrabi, A. Venkat, K. Skadron. *IISWC, 2024.* [link]

[2] **An Efficient Scaling-Free Folded Hyperbolic CORDIC Design Using a Novel Low-Complexity Power-of-2 Taylor Series Approximation**
Anu Verma, **Khyati Kiyawat**, Bishnu P. Das, Pramod K. Meher. *TVLSI, 2023.* [link]

[1] **Real-Time Minimum Energy Point Tracking Using a Predetermined Optimal Voltage Setting Strategy**
Khyati Kiyawat, Yutaka Masuda, Jun Shiomi, Tohru Ishihara. *ISVLSI, 2020.* [link]

Teaching and Mentoring

Graduate Teaching Assistant | University of Virginia, USA
- Computer Systems and Organization I (CS 2130) *Fall 2023*
- Computer Systems and Organization II (CS 3130) *Spring 2024*
- CPU/GPU Memory Systems (CS 6501 Special Topics) *Spring 2025, Spring 2026*

Graduate Mentorship Programs | University of Virginia, USA 2024–present
- Mentoring an undergraduate student on KV cache optimization research for Agentic AI (with Dr. Skadron).
- Mentored a junior PhD student on CS doctoral program navigation (CS Graduate Student Group).
- Mentored a CS undergraduate on graduate school applications (Engineering Grad School Mentor Program).

Rise Together | Charlottesville, Virginia Oct 2023–Apr 2024
- Assisted high-school students in the US with undergraduate college applications and the transition process.

Student Mentorship Programme | IIT Roorkee, India Aug 2018–May 2020
- Mentored 12 first-year students in their transition and adjustment to the institute's culture.

Honors and Awards

- **Most Memorable Internship Project** | Samsung Semiconductors Intern Showcase *Aug 2026*
- **Best Demo Award** | JUMP 2.0 PRISM Annual Review *Nov 2025*
- **Selected Participant** | CRA-WP Grad Cohort Workshop for Women *2023*
- **Engineering Distinguished Fellowship** | University of Virginia, awarded to top Ph.D. applicants *2022*
- **Semi-finalist** | Swadeshi Microprocessor Challenge, RISC-V autonomous distress system *2020–21*
- **Honda Y-E-S Award** | 1 of 14 recipients across 6 IITs; Y-E-S Plus Scholarship *2018–19*
- **Quarter-finalist** | TI India Innovation Challenge Design Contest (IICDC) *2018*
- **1988 Batch Award** | IIT Roorkee, top undergraduate by academic merit *2017*

Service

Artifact Evaluation Committee | ISCA 2024, 2025
- Reviewed research artifacts accompanying accepted papers.

Steering Committee Member | Computer Architecture Student Association (CASA) 2025–present
- Leading the Computer Architecture Long-term Mentorship (CALM) program.

Systems Interest Group (SIG) | University of Virginia Nov 2022–May 2024
- Organized cross-departmental systems research forum, facilitating faculty-student discussions.

Community Outreach | IIT Roorkee, India
- **Unnat Bharat Abhiyan**: Surveyed farming challenges in partner villages under a national program that pairs IITs with rural communities, and designed a sensing system to protect farms from wild animals *(2016–18)*.
- **SOPAN**: Ran weekly outreach activities for primary and middle school students in rural Uttarakhand *(2018–20)*.

Presentations

- Reimagining LLM inference with Memory-centric Servers at *ORNL Core Universities AI Workshop 2026*.
- Open-Source Processing-In-Memory (PIM) Architecture Design through FPGA Emulation: A Case Study Modeling Sieve at *Open-Source Computer Architecture Research Workshop* [OSCAR @ ISCA 2023].
- PIMperf: Analytical Modeling for SIMD Instructions on PIM at *CAPWIC 2024*.
- Chiplet-based DRAM PIM Architecture for LLM Inference at *TECHCON 2024*.